Geometry: Diagnostic G

Time allowed: 55 minutes **Total:** 14 diagnostic marks

Instructions Answer both questions. Show full working and give complete proofs. No calculator, internet, AI assistance or hints during an independent diagnostic. Use separate answer paper if required.

Candidate: _____ Date: _____

Question 1 [7 marks]

GEO-01

In an acute triangle ABC, let E be the foot of the perpendicular from B to AC, and let F be the foot of the perpendicular from C to AB. Let ℓ be the tangent at A to the circumcircle of ABC.

Prove that EF is parallel to ℓ. You may use a clearly labelled diagram.

Question 2 [7 marks]

GEO-02

Let ABC be a nondegenerate triangle, and let P be a point on its circumcircle, distinct from A, B, and C. Let X, Y, Z be the perpendicular projections of P onto the *lines* AB, BC, and CA, respectively. The feet may lie on extensions of the sides.

Prove that X, Y, Z are collinear. Address any coincident-foot cases.

End of paper. Write your answers on separate paper, labelled with the question number.